

Generalized linear models

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Last time: Exponential dispersion models

We say that a distribution belongs to the *exponential dispersion model* family if the distribution's probability function can be written in the form

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

Generalized linear models

For iid observations $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$, a **generalized linear model** (GLM) consists of two components:

$$Y_i | \mathbf{x}_i \sim \text{EDM}(\mu_i, \phi)$$
$$g(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$

Things we might do with a model

$$Y_i | \mathbf{x}_i \sim \text{EDM}(\mu_i, \phi)$$

$$g(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$

- ▶ Estimate $\boldsymbol{\beta}$
- ▶ Model diagnostics
- ▶ Prediction
- ▶ Inference

Activity

Work on the activity (handout).